Hrishit Chaudhuri

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Education

- **Courant Institute of Mathematical Sciences, New York University** — New York, US
  Masters in Computer Science (MSCS); GPA: ***3.895/4.00*** — *Sep 2023 - Present*
- **PES University** — Bangalore, India
  Bachelor of Technology - Computer Science & Engineering; GPA: ***9.59/10.00*** — *Aug 2019 - Jun 2023*
  First Class with Hons. *Specialization:* ***Systems and Core Computing.***

Experience

- **Qualcomm** — San Diego
  Interim Engineering Intern — *May 2024 - Aug 2024*
  - Developed novel MLIR dialect for API to express parallellism constructs for PyTorch front-ends and corresponding LLVM IR lowering mechanism.
- **Qualcomm** — Hyderabad
  Interim Engineering Intern — *Jan 2023 - Jun 2023*
  - Optimized LLVM-based compilers for proprietary microcontrollers, improving performance against benchmarks by 8-10%.
- **Indian Institute of Science - Programming Languages Lab** — Dept of Computer Science & Automation
  Research Intern — *May 2022 - Jul 2022*
  - Integrated neural network verifiers into formal verification framework for cyber-physical systems.

Teaching

- **NYU Courant** — New York City
  Grader — *Sep 2024 - Dec 2024*
  - Grading weekly assignments for graduate-level course on **Programming Languages** by Dr Joseph Tassarotti.
- **NYU Courant** — New York City
  Grader — *Jan 2024 - May 2024*
  - Grading weekly assignments for undergraduate-level course on **Quantum Computing** by Dr Nicholas Spooner.
- **PES University** — Bangalore
  Teaching Assistant — *Aug 2022 - Dec 2022*
  - Designed course material, assignments, and other preparatory instructions for undergraduate-level course on **Advanced Algorithms** by Dr Shobana Padmanabhan.

Technical Projects

- **Certifiably Correct Optimization Passes for Hardware Security Modules** — New York University
  Independent Study | Guide: Dr Joseph Tassarotti — *Sep 2024 - Dec 2024*
  - Developing verification pathways to check correctness for GCC and clang-based optimization pathways using Alive and TVI frameworks to optimize kernels for hardware security modules.
- **ScripTONEs: Automated Film Soundtrack Generation** — PES University
  Capstone Project | Guide: Dr Gowri Srinivasa — *Jan 2022 - Dec 2022*
  - Developed an end-to-end pipeline to generate musical scores for movie scripts based on sentiment conditioning using **variational auto-encoders** and **latent space regularization**.
- **ParCBench: Benchmarking Parallel System Calls to a File System** — PES University
  Course Project | Instructor: Dr KV Subramaniam — *Sep 2022 - Nov 2022*
  - Extended the **lmbench** suite to provide benchmarking for parallel system calls to a file system.
- **ll-SpMV: Fast Highly Parallel Boolean Sparse Matrix Multiplications with OpenMP** — PES University
  Course Project | Instructor: Dr Sudarshan TSB — *Jan 2022 - May 2022*
  - Parallelized the **Four Russians' method** for fast matrix multiplication using **OpenMP**, reducing the run-time by 50%, even with varying sparsity.

Reading Projects

- **Systems of Logic for Verification** — PES University
  Independent Study | Guide: Dr Reetinder Sidhu — *Apr 2021 - Nov 2021*
    - Compiled report on various models of logic used for circuit verification as well as **TLA+**.
    - Identified technical difficulties and limitations in the use of **TLA+** for circuit verification.
- **Boolean Circuit Complexity** — Institute of Mathematical Sciences (Remote)
  Directed Study | Guide: Dr Meena Mahajan — *May 2021 - Jul 2021*
    - Surveyed graduate-level lecture notes on Boolean circuit complexity, discussing advances including **Sipser, Saxe and Furst's proof of** $PARITY \notin AC0$.
    - Assembled notes and summaries to discuss progress at weekly meetings with the guide.

Publications

- **ScripTONES: Sentiment-Conditioned Music Generation for Movie Scripts**: Vishruth Veerendranath, Vibha Masti, Utkarsh Gupta, **Hrishit Chaudhuri**, Gowri Srinivasa. Presented at the Machine Learning for Audio workshop @ **NeurIPS '23** and the Generative AI workshop @ **AI-ML Systems '23.**

Skills Summary

- **Languages**: Python, C, C++, JavaScript, SQL, Java, MATLAB, TLA+, Haskell, Coq, Iris, Go
- **Frameworks**: Tensorflow, PyTorch, OpenMP, CUDA, Pthreads, MLIR, LLVM, Iris
- **Tools**: Git, GitHub, PostgreSQL, Docker, MongoDB, Neo4j

Academic Achievements

- Conferred a scholarship to the **Programming Languages Mentoring Workshop (PLMW)** @ the **Symposium on Principles of Programming Languages (POPL) '23**.
- Awarded the **Dr CNR Rao Scholarship, PES University** across three semesters, in recognition of ranking within the top 2% of students in the batch.
- Awarded the **Dr MR Doreswamy Scholarship, PES University** across five semesters, in recognition of ranking within the top 20% of students in the batch.
- Cracked the final round of **Chennai Mathematical Institute**'s annual examination in the mathematical sciences, in Computer Science, category B (undergraduate level). - January, 2021

Extra-Curricular Activities

- Qualified for **ACM India**'s, in collaboration with the **Chennai Mathematical Institute** and the **Indian Institute of Technology, Madras**, *Winter School on Algorithmic Lower Bounds*, covering topics on fine-grained complexity, between *3rd and 12th January, 2022.*
- Participated in the *International School-conference in Algorithms, Combinatorics, and Complexity*, hosted by the **Department of Mathematics and Computer Science of St Petersburg University** and **Leonhard Euler International Mathematical Institute in Saint Petersburg** between *24th and 28th May, 2021.*
- Reached Stage 2 in *e-Yantra Robotics Competition 2020*, hosted by the **Indian Institute of Technology, Bombay**.
- Senior member and technical staff for **PES Innovation Lab**, PES University's oldest technical club and student-run lab.
- Organized the flagship hackathon **HashCode** for the years 2021 and 2022, and mentored 3 cross-domain projects as part of **PES Innovation Lab**.
- Facilitated the creation of **PES University**'s monthly newsletter, **FYI**, serving as **Chief Editor** for a total of 13 editions across 2 years.
- Served as club head for **Write Angle**, **PES University**'s literary club.
- Interested in Western Classical violin and chess at an amateur level.